Linear Algebra

Week 3

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1 The complex plane, modulus, and argument

We begin with the complex-number topics in the lesson outline: the complex plane, modulus, and argument. Recall that a complex number has the form $z = a + ib$, where $a, b \in \mathbb{R}$ and $i^2 = -1$. Addition and multiplication are given by

$$(a + ib) + (c + id) = (a + c) + i(b + d), \qquad (a + ib)(c + id) = (ac - bd) + i(ad + bc).$$

These formulas let us calculate algebraically. We now give the same numbers a geometric interpretation.

1.1 Real and imaginary parts

Definition 1.1 (Real and imaginary parts)**.** If $z = a + ib$, with $a, b \in \mathbb{R}$, then the *real part* of z is $\mathrm{Re}(z) = a$ and its *imaginary part* is $\mathrm{Im}(z) = b$.

Both $\mathrm{Re}(z)$ and $\mathrm{Im}(z)$ are real numbers. In particular, the imaginary part of $a + ib$ is b, not ib.

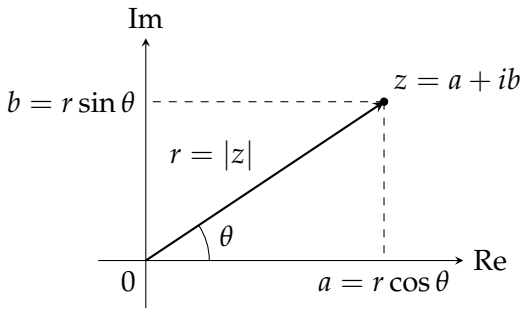
Example 1.1. For $z = 5 - 2i$,

$$\mathrm{Re}(z) = 5, \qquad \mathrm{Im}(z) = -2.$$

For $w = -3i$, we have $\mathrm{Re}(w) = 0$ and $\mathrm{Im}(w) = -3$. For the real number $7 = 7 + 0i$, we have $\mathrm{Re}(7) = 7$ and $\mathrm{Im}(7) = 0$.

A complex number z is real exactly when $\mathrm{Im}(z) = 0$. It is *purely imaginary* when $\mathrm{Re}(z) = 0$, so that $z = i\,\mathrm{Im}(z)$. With this convention, zero is both real and purely imaginary.

We represent $z = a + ib$ by the point (a, b) in the plane. The horizontal axis is the *real axis*, and the vertical axis is the *imaginary axis*. This coordinate plane is called the *complex plane*, or the *Gauss plane*. Thus the algebraic expression $a + ib$ and the point (a, b) are two descriptions of the same complex number.



1.2 The modulus and its geometric meaning

Definition 1.2 (Modulus)**.** The *modulus*, or absolute value, of $z = a + ib$ is

$$|z| = \sqrt{a^2 + b^2}.$$

By the Pythagorean theorem, $|z|$ is the distance from (a, b) to the origin. Consequently,

$$|z| \geq 0, \qquad |z| = 0 \iff z = 0.$$

For real z, this definition agrees with the usual absolute value.

Example 1.2.

$$|-3i| = \sqrt{0^2 + (-3)^2} = 3, \qquad |-6| = \sqrt{(-6)^2 + 0^2} = 6,$$

while

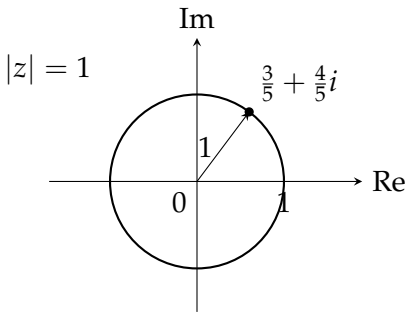
$$|-3+4i| = \sqrt{(-3)^2 + 4^2} = \sqrt{25} = 5.$$

A negative coordinate does not make the modulus negative: a distance is always nonnegative.

Example 1.3 (Circles in the complex plane)**.** The equation $|z| = 1$ describes all points at distance 1 from the origin. Writing $z = a + ib$, we obtain

$$|z| = 1 \iff \sqrt{a^2 + b^2} = 1 \iff a^2 + b^2 = 1.$$

This is the *unit circle*. More generally, $|z| = R$ describes the circle of radius R centered at the origin when $R > 0$. When $R = 0$, it describes only the origin.



Example 1.4 (A distance condition)**.** A sensor position is represented by $z = a + ib$, and a reference point is represented by $2 + i$. The condition

$$|z - (2+i)| = 3$$

means

$$\sqrt{(a-2)^2 + (b-1)^2} = 3, \qquad \text{or equivalently} \qquad (a-2)^2 + (b-1)^2 = 9.$$

Thus the possible positions form a circle of radius 3 centered at $(2, 1)$. This uses the same modulus formula, applied to the displacement between two points.

1.3 The principal argument

Definition 1.3 (Principal argument)**.** For $z \neq 0$, the *principal argument* $\operatorname{Arg}(z)$ is the angle θ from the positive real axis to the ray from the origin through z, chosen in the interval

$$-\pi < \theta \leq \pi.$$

Angles are measured counterclockwise when positive and clockwise when negative.

The argument of 0 is undefined: there is no distinguished ray from the origin to itself. The interval includes π and excludes $-\pi$, so a negative real number has principal argument π.

Example 1.5.

$$\operatorname{Arg}(4i) = \frac{\pi}{2}, \qquad \operatorname{Arg}(-2i) = -\frac{\pi}{2}, \qquad \operatorname{Arg}(-7) = \pi,$$

and

$$\operatorname{Arg}(2-2i) = -\frac{\pi}{4}.$$

Proposition 1.4 (Finding the argument from the coordinates)**.** Let $z = a + ib \neq 0$ and let $\theta = \operatorname{Arg}(z)$. If $a \neq 0$, then

$$\tan\theta = \frac{b}{a}.$$

If $a = 0$, then θ is $\pi/2$ or $-\pi/2$, and $\tan\theta$ is undefined.

Proof. From the right-triangle interpretation in the complex plane,

$$\cos\theta = \frac{a}{|z|}, \qquad \sin\theta = \frac{b}{|z|}.$$

If $a \neq 0$, division gives

$$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{b/|z|}{a/|z|} = \frac{b}{a}.$$

If $a = 0$, the point lies on the imaginary axis, so its nonzero imaginary coordinate determines which of the two angles applies. □

The tangent alone does not determine the argument: angles differing by π have the same tangent. We must also use the quadrant.

Example 1.6 (The quadrant matters)**.** Take $z = -\sqrt{3} + i$. Then

$$\tan\theta = \frac{1}{-\sqrt{3}} = -\frac{1}{\sqrt{3}}.$$

The number lies in the second quadrant, so

$$\theta = \frac{5\pi}{6},$$

not $-\pi/6$. In contrast, $w = \sqrt{3} - i$ has the same tangent ratio but lies in the fourth quadrant, so $\operatorname{Arg}(w) = -\pi/6$.

Example 1.7. For $z = -1 - \sqrt{3}i$, the tangent ratio is $\sqrt{3}$. The point is below the real axis and to the left of the imaginary axis. Its principal argument is therefore

$$\operatorname{Arg}(z) = -\frac{2\pi}{3}.$$

Although $4\pi/3$ describes the same ray, it lies outside the chosen interval.

2 Polar form and complex arithmetic

2.1 Passing between Cartesian and polar form

Theorem 2.1 (Polar representation)**.** Every nonzero complex number has a representation

$$z = r(\cos\theta + i\sin\theta), \qquad r = |z| > 0, \qquad \theta = \operatorname{Arg}(z).$$

For $z = 0$, the same expression equals 0 when $r = 0$, whatever the value of θ.

Proof. Let $z = a + ib \neq 0$ and $r = \sqrt{a^2 + b^2}$. The point $(a/r, b/r)$ lies on the unit circle because

$$\left(\frac{a}{r}\right)^2 + \left(\frac{b}{r}\right)^2 = \frac{a^2 + b^2}{r^2} = 1.$$

Its angle θ therefore satisfies $\cos\theta = a/r$ and $\sin\theta = b/r$. Multiplying by r gives

$$r(\cos\theta + i\sin\theta) = a + ib = z.$$

□

Cartesian form uses the two coordinates a, b. Polar form uses the radius r and the angle θ. The conversion rules are

$$a = r\cos\theta, \qquad b = r\sin\theta$$

in one direction, and

$$r = \sqrt{a^2 + b^2}, \qquad \tan\theta = \frac{b}{a} \quad (a \neq 0),$$

with the quadrant determining the correct angle, in the other direction.

Example 2.1. For $z = 2\sqrt{3} - 2i$,

$$r = \sqrt{(2\sqrt{3})^2 + (-2)^2} = \sqrt{12+4} = 4, \qquad \operatorname{Arg}(z) = -\frac{\pi}{6}.$$

Hence

$$2\sqrt{3} - 2i = 4\left(\cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right)\right).$$

Conversely,

$$\begin{aligned} 3\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right) &= 3\left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) \\ &= -\frac{3}{2} + \frac{3\sqrt{3}}{2}i. \end{aligned}$$

2.2 When two polar expressions are equal

Because sine and cosine are 2π-periodic, a polar expression need not use the principal argument. For example,

$$2i = 2\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right) = 2\left(\cos\frac{5\pi}{2} + i\sin\frac{5\pi}{2}\right).$$

Theorem 2.2 (Uniqueness of polar form)**.** Suppose $r, s > 0$. Then

$$r(\cos\theta + i\sin\theta) = s(\cos\varphi + i\sin\varphi)$$

if and only if

$$r = s \qquad \text{and} \qquad \theta - \varphi = 2k\pi \quad \text{for some } k \in \mathbb{Z}.$$

Proof. Taking moduli gives $r = s$, since $\cos^2 t + \sin^2 t = 1$. Dividing by this common nonzero radius and comparing real and imaginary parts gives

$$\cos\theta = \cos\varphi, \qquad \sin\theta = \sin\varphi.$$

The angle-difference identities imply

$$\cos(\theta - \varphi) = \cos\theta\cos\varphi + \sin\theta\sin\varphi = 1,$$

and

$$\sin(\theta - \varphi) = \sin\theta\cos\varphi - \cos\theta\sin\varphi = 0.$$

Thus $\theta - \varphi = 2k\pi$ for an integer k. Conversely, such a difference leaves both sine and cosine unchanged. □

If both angles are required to lie in $(-\pi, \pi]$, then they are equal. When $r = 0$, every angle gives the same number 0, so the nonzero-radius condition is essential.

2.3 Multiplication: multiply radii and add angles

Theorem 2.3 (Multiplication in polar form)**.** If

$$z_1 = r_1(\cos\theta_1 + i\sin\theta_1), \qquad z_2 = r_2(\cos\theta_2 + i\sin\theta_2),$$

where $r_1, r_2 \geq 0$, then

$$z_1 z_2 = r_1 r_2\big(\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)\big).$$

Proof. Expand the product and use $i^2 = -1$:

$$\begin{aligned} z_1 z_2 &= r_1 r_2(\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2) \\ &= r_1 r_2\big[(\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2) \\ &\qquad\qquad + i(\sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2)\big] \\ &= r_1 r_2\big(\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)\big). \end{aligned}$$

The last line is precisely the pair of trigonometric angle-addition identities. □

Example 2.2 (The same product in two forms)**.** Let

$$z_1 = \sqrt{3} + i = 2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right), \qquad z_2 = 1 + \sqrt{3}i = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right).$$

The radii multiply to 4, and the angles add to $\pi/2$. Therefore

$$z_1 z_2 = 4\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right) = 4i.$$

The Cartesian calculation confirms this:

$$(\sqrt{3}+i)(1+\sqrt{3}i) = \sqrt{3} + 3i + i - \sqrt{3} = 4i.$$

Example 2.3 (A rotation without changing the distance)**.** Multiplication by i sends

$$a + ib \longmapsto i(a+ib) = -b + ia.$$

Thus (a, b) becomes $(-b, a)$: a counterclockwise rotation by $\pi/2$. For example,

$$i(3-2i) = 2 + 3i.$$

Both points are at distance $\sqrt{13}$ from the origin:

$$|3-2i| = \sqrt{9+4} = \sqrt{13}, \qquad |2+3i| = \sqrt{4+9} = \sqrt{13}.$$

More generally, multiplying by $\cos\alpha + i\sin\alpha$ rotates by α because its radius is 1.

Addition is usually simpler in Cartesian form, where coordinates add separately. Multiplication is often simpler in polar form, where radii multiply and angles add.

Remark 2.4 (A polar angle need not be the principal argument)**.** For nonzero factors, it is not always true that $\operatorname{Arg}(z_1 z_2) = \operatorname{Arg}(z_1) + \operatorname{Arg}(z_2)$ as an equality of real numbers. The sum may lie outside $(-\pi, \pi]$. It is an angle for the product, and differs from the principal argument by an integer multiple of 2π.

Example 2.4. If $z_1 = z_2 = -1 + i$, each factor has principal argument $3\pi/4$. The sum is $3\pi/2$, whereas

$$(-1+i)^2 = 1 - 2i + i^2 = -2i, \qquad \operatorname{Arg}(-2i) = -\frac{\pi}{2}.$$

The two angles differ by 2π, so they give the same polar expression.

2.4 Reciprocals and division

Theorem 2.5 (Reciprocal and quotient in polar form)**.** If $z = r(\cos\theta + i\sin\theta) \neq 0$, then

$$z^{-1} = \frac{1}{r}\big(\cos(-\theta) + i\sin(-\theta)\big).$$

Consequently, if $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2) \neq 0$, then

$$\frac{z_1}{z_2} = \frac{r_1}{r_2}\big(\cos(\theta_1 - \theta_2) + i\sin(\theta_1 - \theta_2)\big).$$

Proof. The proposed reciprocal has radius $1/r$ and angle $-\theta$. By the multiplication formula, its product with z is

$$r\frac{1}{r}\big(\cos(\theta - \theta) + i\sin(\theta - \theta)\big) = \cos 0 + i\sin 0 = 1.$$

This proves that it is an inverse. Multiplying z_1 by the reciprocal of z_2 gives the quotient formula. □

Example 2.5. The numerator $\sqrt{3} + i$ has radius 2 and angle $\pi/6$, while the denominator $1 + i$ has radius $\sqrt{2}$ and angle $\pi/4$. Hence

$$\begin{aligned}\frac{\sqrt{3}+i}{1+i} &= \frac{2}{\sqrt{2}}\left(\cos\left(\frac{\pi}{6} - \frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{6} - \frac{\pi}{4}\right)\right)\\ &= \sqrt{2}\left(\cos\left(-\frac{\pi}{12}\right) + i\sin\left(-\frac{\pi}{12}\right)\right).\end{aligned}$$

As with multiplication, the difference of the input angles is a valid polar angle even if it is outside the principal interval.

2.5 Powers and De Moivre's formula

Theorem 2.6 (De Moivre's formula)**.** If $z = r(\cos\theta + i\sin\theta)$, then for every positive integer n,

$$z^n = r^n\big(\cos(n\theta) + i\sin(n\theta)\big).$$

If $z \neq 0$, the identity also holds for every integer n, including 0 and negative integers.

Proof. For $n = 1$, the identity is the given representation of z. Suppose it holds for some positive integer n. Multiplying by z and using the polar multiplication formula gives

$$\begin{aligned}z^{n+1} &= r^n\big(\cos(n\theta) + i\sin(n\theta)\big)r(\cos\theta + i\sin\theta)\\ &= r^{n+1}\big(\cos((n+1)\theta) + i\sin((n+1)\theta)\big).\end{aligned}$$

Induction proves the result for all positive integers. For $z \neq 0$, the case $n = 0$ reads $1 = \cos 0 + i\sin 0$. If $n = -m$ with $m > 0$, apply the positive-power formula to

$$z^{-1} = \frac{1}{r}\big(\cos(-\theta) + i\sin(-\theta)\big)$$

to obtain

$$z^{-m} = r^{-m}\big(\cos(-m\theta) + i\sin(-m\theta)\big).$$

□

Example 2.6 (A high power with little expansion)**.** Since $\sqrt{3} + i = 2(\cos(\pi/6) + i\sin(\pi/6))$,

$$\begin{aligned}(\sqrt{3}+i)^6 &= 2^6\left(\cos\frac{6\pi}{6} + i\sin\frac{6\pi}{6}\right)\\ &= 64(\cos\pi + i\sin\pi) = -64.\end{aligned}$$

The sixth power is real, even though the original number is not real.

Example 2.7 (Repeated rotations return to the starting point)**.** Let $u = \frac{\sqrt{3}}{2} + \frac{1}{2}i$. It has radius 1 and angle $\pi/6$, so multiplication by u rotates through 30 degrees. After twelve multiplications,

$$u^{12} = \cos(2\pi) + i\sin(2\pi) = 1.$$

For a concrete point $z = 2 - i$, this means $u^{12}z = z$. After six multiplications, $u^6 = -1$, so $u^6 z = -2 + i$, the point opposite z across the origin.

3 Conjugation and properties of complex numbers

3.1 Complex conjugation

Definition 3.1 (Complex conjugate)**.** For $z = a + ib$, the *complex conjugate* of z is

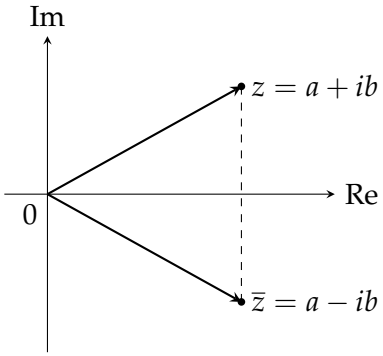
$$\overline{z} = a - ib.$$

Geometrically, conjugation reflects a point across the real axis. The real coordinate stays the same, while the imaginary coordinate changes sign.

Example 3.1.

$$\overline{2 - 5i} = 2 + 5i, \qquad \overline{-4 + 3i} = -4 - 3i, \qquad \overline{-7} = -7, \qquad \overline{-6i} = 6i.$$

Conjugation leaves real numbers fixed. For purely imaginary numbers, it has the same effect as multiplication by -1.



Proposition 3.2 (Conjugation in polar form)**.** If $z = r(\cos\theta + i\sin\theta)$, then

$$\overline{z} = r\big(\cos(-\theta) + i\sin(-\theta)\big).$$

Proof. Conjugation changes $r\cos\theta + ir\sin\theta$ to $r\cos\theta - ir\sin\theta$. Since $\cos(-\theta) = \cos\theta$ and $\sin(-\theta) = -\sin\theta$, this is the displayed polar expression. □

Example 3.2. The conjugate of $4(\cos(\pi/3) + i\sin(\pi/3)) = 2 + 2\sqrt{3}i$ is

$$4\left(\cos\left(-\frac{\pi}{3}\right) + i\sin\left(-\frac{\pi}{3}\right)\right) = 2 - 2\sqrt{3}i.$$

3.2 The basic identities

Theorem 3.3 (Properties of complex numbers)**.** For $z, z_1, z_2 \in \mathbb{C}$, the following identities and inequalities hold. Quotient identities require $z_2 \neq 0$.

1. $\operatorname{Re}(z_1 + z_2) = \operatorname{Re}(z_1) + \operatorname{Re}(z_2)$ and $\operatorname{Im}(z_1 + z_2) = \operatorname{Im}(z_1) + \operatorname{Im}(z_2)$.
2. $\overline{\overline{z}} = z$.
3. $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$.
4. $\overline{z_1 z_2} = \overline{z_1}\,\overline{z_2}$.
5. $\overline{\left(\dfrac{z_1}{z_2}\right)} = \dfrac{\overline{z_1}}{\overline{z_2}}$.
6. $z + \overline{z} = 2\operatorname{Re}(z)$.
7. $z - \overline{z} = 2i\operatorname{Im}(z)$.
8. $|\operatorname{Re}(z)| \leq |z|$ and $|\operatorname{Im}(z)| \leq |z|$.
9. $z\overline{z} = |z|^2$.
10. $|\overline{z}| = |z|$.
11. $|z_1 z_2| = |z_1|\,|z_2|$.
12. $\left|\dfrac{z_1}{z_2}\right| = \dfrac{|z_1|}{|z_2|}$.
13. $|z_1 + z_2| \leq |z_1| + |z_2|$ *(the triangle inequality)*.

Proof. Write $z = a + ib$ and $z_j = a_j + ib_j$, with all coordinates real.

For the first identity, addition gives

$$z_1 + z_2 = (a_1 + a_2) + i(b_1 + b_2),$$

so its real and imaginary parts are $a_1 + a_2$ and $b_1 + b_2$. Conjugating twice changes b to $-b$ and then back to b, proving the second identity.

For conjugation of a sum,

$$\begin{aligned}\overline{z_1 + z_2} &= (a_1 + a_2) - i(b_1 + b_2) \\ &= (a_1 - ib_1) + (a_2 - ib_2) = \overline{z_1} + \overline{z_2}.\end{aligned}$$

For conjugation of a product, first calculate

$$z_1 z_2 = (a_1 a_2 - b_1 b_2) + i(a_1 b_2 + b_1 a_2).$$

Therefore

$$\overline{z_1 z_2} = (a_1 a_2 - b_1 b_2) - i(a_1 b_2 + b_1 a_2).$$

On the other hand,

$$\begin{aligned}\overline{z_1}\,\overline{z_2} &= (a_1 - ib_1)(a_2 - ib_2) \\ &= (a_1 a_2 - b_1 b_2) - i(a_1 b_2 + b_1 a_2),\end{aligned}$$

which is the same result.

If $z_2 \neq 0$, conjugating the identity $z_2 z_2^{-1} = 1$ gives

$$\overline{z_2}\,\overline{z_2^{-1}} = 1.$$

Thus $\overline{z_2^{-1}} = (\overline{z_2})^{-1}$, and hence

$$\overline{\left(\frac{z_1}{z_2}\right)} = \overline{z_1 z_2^{-1}} = \overline{z_1}\,(\overline{z_2})^{-1} = \frac{\overline{z_1}}{\overline{z_2}}.$$

The next two identities follow by adding and subtracting:

$$z + \overline{z} = (a + ib) + (a - ib) = 2a = 2\,\mathrm{Re}(z),$$

$$z - \overline{z} = (a + ib) - (a - ib) = 2ib = 2i\,\mathrm{Im}(z).$$

For the coordinate bounds, use

$$|\,\mathrm{Re}(z)| = \sqrt{a^2} \le \sqrt{a^2 + b^2} = |z|, \qquad |\,\mathrm{Im}(z)| = \sqrt{b^2} \le \sqrt{a^2 + b^2} = |z|.$$

The conjugate product is

$$z\overline{z} = (a + ib)(a - ib) = a^2 - i^2 b^2 = a^2 + b^2 = |z|^2.$$

Also,

$$|\overline{z}| = \sqrt{a^2 + (-b)^2} = \sqrt{a^2 + b^2} = |z|.$$

To obtain multiplicativity of the modulus, use the product identity already proved:

$$\begin{aligned} |z_1 z_2|^2 &= (z_1 z_2)\overline{(z_1 z_2)} \\ &= z_1 z_2 \overline{z_1}\,\overline{z_2} \\ &= (z_1 \overline{z_1})(z_2 \overline{z_2}) = |z_1|^2 |z_2|^2. \end{aligned}$$

Both moduli are nonnegative, so taking square roots yields $|z_1 z_2| = |z_1||z_2|$. For the quotient, $z_1 = (z_1/z_2)z_2$ implies

$$|z_1| = \left|\frac{z_1}{z_2}\right| |z_2|.$$

Since $|z_2| > 0$, divide by it to obtain the quotient formula.

Finally,

$$\begin{aligned} |z_1 + z_2|^2 &= (z_1 + z_2)(\overline{z_1} + \overline{z_2}) \\ &= |z_1|^2 + |z_2|^2 + z_1\overline{z_2} + \overline{z_1} z_2 \\ &= |z_1|^2 + |z_2|^2 + 2\,\mathrm{Re}(z_1 \overline{z_2}). \end{aligned}$$

The real-part bound and multiplicativity give

$$\mathrm{Re}(z_1 \overline{z_2}) \le |z_1 \overline{z_2}| = |z_1||\overline{z_2}| = |z_1||z_2|.$$

Consequently,

$$|z_1 + z_2|^2 \le |z_1|^2 + 2|z_1||z_2| + |z_2|^2 = (|z_1| + |z_2|)^2.$$

Taking nonnegative square roots proves the triangle inequality. □

Example 3.3 (Two paths and one displacement)**.** Let $z_1 = 3 + 4i$ and $z_2 = -3 + 4i$. Each displacement has length 5, while their sum is $8i$. Thus

$$|z_1 + z_2| = 8 < 5 + 5 = |z_1| + |z_2|.$$

Following the two displacements has total length 10, while going directly from the starting point to the final point has length 8. This is the geometric meaning of the triangle inequality.

3.3 Division using the conjugate

Corollary 3.4. For $z \neq 0$,

$$z^{-1} = \frac{\overline{z}}{|z|^2}.$$

More generally, for $z_2 \neq 0$,

$$\frac{z_1}{z_2} = \frac{z_1 \overline{z_2}}{|z_2|^2}.$$

Proof. The identity $z\overline{z} = |z|^2$ and the fact $|z|^2 > 0$ give

$$z\left(\frac{\overline{z}}{|z|^2}\right) = \frac{z\overline{z}}{|z|^2} = 1.$$

This proves the reciprocal formula. Multiplying by z_1 proves the quotient formula. □

Multiplying numerator and denominator by the denominator's conjugate replaces a complex denominator by a positive real one.

Example 3.4. To express $(5+2i)/(1-2i)$ in Cartesian form, calculate

$$\begin{aligned}\frac{5+2i}{1-2i} &= \frac{(5+2i)(1+2i)}{(1-2i)(1+2i)} \\ &= \frac{5+10i+2i+4i^2}{1+4} \\ &= \frac{1+12i}{5} = \frac{1}{5} + \frac{12}{5}i.\end{aligned}$$

As a check, multiplying the answer by $1-2i$ gives

$$\begin{aligned}\left(\frac{1}{5} + \frac{12}{5}i\right)(1-2i) &= \frac{1}{5} - \frac{2}{5}i + \frac{12}{5}i - \frac{24}{5}i^2 \\ &= 5 + 2i,\end{aligned}$$

the original numerator.

3.4 The complex numbers form a field

Theorem 3.5. With the addition and multiplication defined above, $\mathbb{C}$ is a field.

Proof. The sum and product of two numbers of the form $a+ib$ again have that form, so both operations are closed in $\mathbb{C}$. Addition is associative and commutative because the real and imaginary coordinates add separately using real addition. The additive identity is $0+0i$, and the additive inverse of $a+ib$ is $-a-ib$.

Multiplication is commutative because

$$(ac-bd) + i(ad+bc) = (ca-db) + i(cb+da).$$

The multiplicative identity is $1+0i$. To check associativity explicitly, let $z = a+ib$, $w = c+id$, and $v = e+if$. Then

$$\begin{aligned}(zw)v &= [(ac-bd) + i(ad+bc)](e+if) \\ &= (ace - bde - adf - bcf) \\ &\quad + i(acf - bdf + ade + bce),\end{aligned}$$

whereas

$$\begin{aligned} z(wv) &= (a+ib)[(ce-df)+i(cf+de)] \\ &= (ace-adf-bcf-bde) \\ &\quad + i(acf+ade+bce-bdf). \end{aligned}$$

The corresponding real and imaginary parts agree, so $(zw)v = z(wv)$.

For distributivity,

$$\begin{aligned} z(w+v) &= (a+ib)[(c+e)+i(d+f)] \\ &= \big(a(c+e)-b(d+f)\big) \\ &\quad + i\big(a(d+f)+b(c+e)\big) \\ &= \big((ac-bd)+i(ad+bc)\big) \\ &\quad + \big((ae-bf)+i(af+be)\big) \\ &= zw+zv. \end{aligned}$$

The other distributive identity follows from commutativity of multiplication.

Finally, $0 \neq 1$, and every nonzero z has the reciprocal $\overline{z}/|z|^2$ proved above. These are exactly the field axioms. □

Preparing for polynomials

Before continuing, review the degree of a polynomial, the meaning of a root, polynomial division, the quadratic Vieta formulas, and the polar form and power formula developed here. These will connect the geometry of complex numbers with the polynomial topics that follow.

4 Polynomials over the complex numbers

We now use complex numbers to understand polynomial equations. The main change from real arithmetic is that a polynomial may have roots outside the real line. The coefficients still matter: real coefficients force non-real roots to occur in conjugate pairs.

4.1 Polynomial notation and multiplicity

Definition 4.1 (Polynomial over a field)**.** Let $\mathbb{F}$ be a field. A polynomial of degree n over $\mathbb{F}$ is an expression

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0, \qquad a_0, \ldots, a_n \in \mathbb{F}, \quad a_n \neq 0.$$

The symbol x is a variable. The set $\mathbb{F}[x]$ contains the zero polynomial and all polynomials of all degrees over $\mathbb{F}$. A nonzero constant polynomial has degree zero. We do not assign an ordinary nonnegative degree to the zero polynomial.

A number $\alpha \in \mathbb{F}$ is a *root* if $f(\alpha) = 0$. We use the factor theorem: $f(\alpha) = 0$ if and only if $x - \alpha$ divides $f(x)$. A root has *multiplicity* m if

$$f(x) = (x - \alpha)^m g(x), \qquad g(\alpha) \neq 0.$$

For example, $(x-3)^2(x+1)$ has root 3 of multiplicity two and root -1 of multiplicity one. Repeated roots must be counted repeatedly when a theorem says "including multiplicity."

4.2 Conjugate roots of real polynomials

Theorem 4.2 (Conjugate-root theorem)**.** If $f \in \mathbb{R}[x]$ is nonzero and $\alpha \in \mathbb{C}$ is a root of f, then $\overline{\alpha}$ is also a root, with the same multiplicity.

Proof. Since every coefficient a_k is real,

$$f(\overline{\alpha}) = \sum_{k=0}^{n} a_k \overline{\alpha}^k = \overline{\sum_{k=0}^{n} a_k \alpha^k} = \overline{f(\alpha)} = 0.$$

To check multiplicity, suppose $f(x) = (x - \alpha)^m g(x)$ and $g(\alpha) \neq 0$. Let $\overline{g}$ denote the polynomial obtained by conjugating every coefficient of g. Conjugating the coefficients of the factorization gives

$$f(x) = (x - \overline{\alpha})^m \overline{g}(x), \qquad \overline{g}(\overline{\alpha}) = \overline{g(\alpha)} \neq 0.$$

Thus the multiplicity is exactly m for both roots. □

Example 4.1 (A conjugate pair)**.** For $f(x) = x^2 - 2x + 5$, the quadratic formula gives

$$x = \frac{2 \pm \sqrt{4 - 20}}{2} = \frac{2 \pm 4i}{2} = 1 \pm 2i.$$

Consequently,

$$x^2 - 2x + 5 = (x - (1 + 2i))(x - (1 - 2i)).$$

Expanding the right-hand side gives $(x-1)^2 - (2i)^2 = x^2 - 2x + 5$, as required.

Remark 4.3. Real coefficients are essential. The polynomial $x - i$ has the root i, but $-i$ is not a root because $(-i) - i = -2i \neq 0$.

4.3 The fundamental theorem of algebra

Theorem 4.4 (Fundamental theorem of algebra)**.** Every polynomial in $\mathbb{C}[x]$ of degree at least one has a complex root.

A field with this property is called *algebraically closed*; in particular, $\mathbb{C}$ is algebraically closed.

Corollary 4.5 (Complete factorization over $\mathbb{C}$)**.** If $f \in \mathbb{C}[x]$ has degree $n \geq 1$ and leading coefficient a_n, then there are $\alpha_1, \ldots, \alpha_n \in \mathbb{C}$ such that

$$f(x) = a_n(x - \alpha_1) \cdots (x - \alpha_n).$$

Thus f has exactly n roots when multiplicity is included.

Proof. The fundamental theorem gives a root α_1. The factor theorem gives $f(x) = (x - \alpha_1)g(x)$, where g has degree $n - 1$ and leading coefficient a_n. Apply the same argument to g and continue until only the constant a_n remains. The degrees decrease by one at each step, so exactly n linear factors occur. □

Remark 4.6 (The quadratic formula over $\mathbb{C}$)**.** For $ax^2 + bx + c$ with $a \neq 0$, the roots are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Here a square root of the complex discriminant may be used; the two choices give the two signs in the formula. General formulas by radicals exist for degrees three and four. There is no formula by radicals that solves every polynomial of degree five or higher.

Example 4.2 (Factoring a cubic)**.** Solve $x^3 + 8 = 0$ over $\mathbb{C}$. Since $(-2)^3 + 8 = 0$, division by $x + 2$ gives

$$x^3 + 8 = (x + 2)(x^2 - 2x + 4).$$

For the quadratic factor,

$$x = \frac{2 \pm \sqrt{4 - 16}}{2} = \frac{2 \pm 2\sqrt{3}\,i}{2} = 1 \pm \sqrt{3}\,i.$$

Therefore

$$x^3 + 8 = (x + 2)(x - 1 - \sqrt{3}\,i)(x - 1 + \sqrt{3}\,i).$$

We have found all three roots, including their multiplicities.

Corollary 4.7 (A real root in odd degree)**.** Every real polynomial of odd degree has a real root.

Proof. Count all its complex roots with multiplicity. Every non-real root is paired with its conjugate, with the same multiplicity, so the number of non-real roots is even. Since the total degree is odd, at least one root is real. □

Example 4.3 (A finite field need not be algebraically closed)**.** Consider $f(x) = x^2 + 1$ over $\mathbb{Z}_7$. The arithmetic below is modulo 7:

x	0	1	2	3	4	5	6
x^2	0	1	4	2	2	4	1
$x^2 + 1$	1	2	5	3	3	5	2

None of the values is zero. Thus f has no root in $\mathbb{Z}_7$, even though $\mathbb{Z}_7$ is a field. This shows that $\mathbb{Z}_7$ is not algebraically closed.

4.4 Vieta's formulas

If $f(x) = a_2(x-\alpha_1)(x-\alpha_2)$, expanding gives

$$f(x) = a_2\big(x^2 - (\alpha_1+\alpha_2)x + \alpha_1\alpha_2\big).$$

Comparing coefficients yields

$$\alpha_1+\alpha_2 = -\frac{a_1}{a_2}, \qquad \alpha_1\alpha_2 = \frac{a_0}{a_2}.$$

The same idea works in every degree.

Theorem 4.8 (Vieta's sum and product formulas)**.** Let $f(x) = a_nx^n + \cdots + a_0 \in \mathbb{C}[x]$, where $n \geq 1$ and $a_n \neq 0$, and let $\alpha_1, \ldots, \alpha_n$ be all its roots, including multiplicity. Then

$$\alpha_1 + \cdots + \alpha_n = -\frac{a_{n-1}}{a_n}, \qquad \alpha_1 \cdots \alpha_n = (-1)^n\frac{a_0}{a_n}.$$

The formulas also hold over any field in which the polynomial splits into linear factors.

Proof. Expand $a_n\prod_{j=1}^{n}(x-\alpha_j)$. To obtain x^{n-1}, choose $-\alpha_j$ from exactly one factor and x from all others. Its coefficient is therefore $-a_n\sum_j \alpha_j$. The constant term is $a_n\prod_j(-\alpha_j) = a_n(-1)^n\prod_j\alpha_j$. Compare these with a_{n-1} and a_0, respectively. □

Example 4.4 (Coefficients from one repeated complex root)**.** Suppose

$$f(x) = x^5 + ax^3 + bx^2 + cx + d \in \mathbb{R}[x]$$

and $1+2i$ is a root of multiplicity two. Find a, b, c, d.

The conjugate $1-2i$ is also a root of multiplicity two. Denote the fifth root by β. Since the coefficient of x^4 is zero, Vieta's sum formula gives

$$2(1+2i) + 2(1-2i) + \beta = 0, \qquad 4+\beta = 0, \qquad \beta = -4.$$

The polynomial is monic, so

$$\begin{aligned} f(x) &= (x-(1+2i))^2(x-(1-2i))^2(x+4) \\ &= (x^2-2x+5)^2(x+4) \\ &= (x^4-4x^3+14x^2-20x+25)(x+4) \\ &= x^5 - 2x^3 + 36x^2 - 55x + 100. \end{aligned}$$

Hence $a=-2$, $b=36$, $c=-55$, and $d=100$. As a check, the product of the five roots is $5^2(-4) = -100$, consistent with $(-1)^5d = -100$.

4.5 Roots of unity

Definition 4.9. Let $n \geq 1$ be an integer. An *nth root of unity* is a complex number z satisfying $z^n = 1$.

For example, 1 and -1 are the square roots of unity, and $1, i, -1, -i$ are the fourth roots of unity. "An nth root of unity" does not require that n be the smallest positive exponent giving 1.

Theorem 4.10 (All roots of unity)**.** There are exactly n distinct nth roots of unity:

$$z_k = \cos\left(\frac{2\pi k}{n}\right) + i\sin\left(\frac{2\pi k}{n}\right), \qquad k = 0, 1, \ldots, n-1.$$

Proof. Write $z = r(\cos\theta + i\sin\theta)$. The equation $z^n = 1$ implies $r^n = 1$, so $r = 1$. De Moivre's formula now gives

$$\cos(n\theta) + i\sin(n\theta) = 1,$$

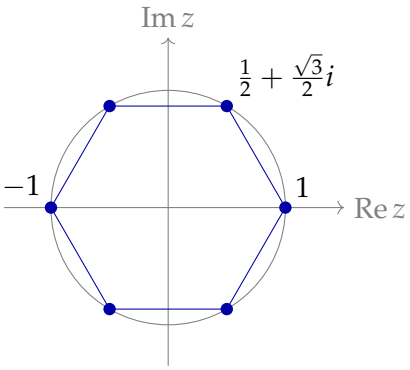
which holds exactly when $n\theta = 2\pi k$ for an integer k. Thus $\theta = 2\pi k/n$. Two such angles represent the same number exactly when their indices differ by a multiple of n, so $k = 0, \dots, n-1$ gives exactly n distinct roots. □

Remark 4.11 (Geometry and sum)**.** The roots lie at equal angular intervals on the unit circle. For $n \geq 2$, their sum is zero: they are the roots of $x^n - 1$, whose x^{n-1} coefficient is zero, so Vieta's formula applies. The condition $n \geq 2$ matters: the sole first root of unity is 1.

Example 4.5 (Six equally spaced points)**.** For $n = 6$, the roots are

$$1, \quad \frac{1}{2} + \frac{\sqrt{3}}{2}i, \quad -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \quad -1, \quad -\frac{1}{2} - \frac{\sqrt{3}}{2}i, \quad \frac{1}{2} - \frac{\sqrt{3}}{2}i.$$

They are the vertices of a regular hexagon. Opposite vertices cancel in pairs, which also makes their sum visibly zero.



4.6 Roots of a general complex number

Theorem 4.12 (nth roots of a nonzero complex number)**.** Let $n \geq 1$ and write

$$w = \rho(\cos\varphi + i\sin\varphi) \neq 0, \qquad \rho > 0.$$

The nth roots of w are exactly

$$z_k = \sqrt[n]{\rho}\left(\cos\left(\frac{\varphi + 2\pi k}{n}\right) + i\sin\left(\frac{\varphi + 2\pi k}{n}\right)\right), \quad k = 0, \dots, n-1.$$

Proof. If $z = r(\cos\theta + i\sin\theta)$ and $z^n = w$, then equality of moduli gives $r^n = \rho$, so $r = \sqrt[n]{\rho}$. Equality of the polar angles gives $n\theta = \varphi + 2\pi k$. The n indices displayed give distinct roots, and every integer index is equivalent to one of them modulo n. □

Example 4.6 (A square-root calculation)**.** Find the solutions of $z^2 = -8i$. Here $\rho = 8$ and $\varphi = -\pi/2$, so the two root angles are $-\pi/4$ and $3\pi/4$. Their common modulus is $\sqrt{8} = 2\sqrt{2}$. Therefore

$$z_0 = 2 - 2i, \qquad z_1 = -2 + 2i.$$

Indeed, $(2 - 2i)^2 = 4 - 8i - 4 = -8i$, and the negative of a square root has the same square.

Example 4.7 (The same cubic, now using geometry)**.** For $z^3 = -8$, use $\rho = 8$, $\varphi = \pi$, and $\sqrt[3]{8} = 2$. The angles are $\pi/3$, π, and $5\pi/3$, giving

$$z_0 = 1 + \sqrt{3}\,i, \qquad z_1 = -2, \qquad z_2 = 1 - \sqrt{3}\,i.$$

These agree with the factorization of $x^3 + 8$ above. Algebraic factorization and polar coordinates describe the same three roots.

If $w = 0$, the only solution of $z^n = 0$ is $z = 0$; its multiplicity as a root of x^n is n.

5 Matrices and their basic notation

A matrix records many scalars in a rectangular array. The positions of its entries are part of the data: a row and a column have different roles. Later, matrix arithmetic will encode systems of equations and compositions of linear transformations.

5.1 Entries, dimensions, and equality

Definition 5.1 (Matrix over a field)**.** An $m \times n$ matrix over $\mathbb{F}$ is a rectangular array with m rows, n columns, and entries in $\mathbb{F}$:

$$A = (a_{ij}) = A_{m\times n} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

The entry in row i and column j is denoted a_{ij} or $(A)_{ij}$. There are mn entries. Round or square brackets may be used.

Example 5.1 (Different shapes and fields)**.** The matrices

$$A = \begin{pmatrix} 2 & 0 & -1 \\ 3 & 5 & \frac{1}{2} \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & i \\ 2-i & 0 \\ -3 & 1+i \end{pmatrix}$$

have sizes 2×3 and 3×2, respectively. The first may be regarded as a matrix over $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$; the second is over $\mathbb{C}$. In A, the entries $a_{12} = 0$ and $a_{21} = 3$ are different positions. The entry a_{32} is undefined, since A has no third row.

A matrix may have only one entry: (9) is a 1×1 matrix. We may identify it with its single scalar when this causes no confusion, but keep the brackets when its matrix role matters.

Definition 5.2 (Equality)**.** Two matrices are equal precisely when they have the same dimensions and equal entries in every corresponding position. Thus

$$A_{m\times n} = B_{r\times s} \iff m = r,\ n = s,\ a_{ij} = b_{ij} \text{ for all } i, j.$$

Example 5.2 (Reading equality entry by entry)**.** Over $\mathbb{R}$,

$$\begin{pmatrix} u+1 & 2 \\ -3 & v-2 \end{pmatrix} = \begin{pmatrix} 5 & 2 \\ -3 & 7 \end{pmatrix} \iff u = 4,\ v = 9.$$

A 2×3 matrix cannot equal a 3×2 matrix, even if their entries list the same six numbers.

5.2 Transpose

Definition 5.3 (Transpose)**.** For $A_{m\times n}$, its *transpose* A^{T} is the $n \times m$ matrix defined by

$$(A^{\mathsf{T}})_{ij} = a_{ji}.$$

In words, rows become columns and columns become rows. The notation A^t is also common.

Example 5.3. For the two matrices above,

$$A^{\mathsf{T}} = \begin{pmatrix} 2 & 3 \\ 0 & 5 \\ -1 & \frac{1}{2} \end{pmatrix}, \qquad B^{\mathsf{T}} = \begin{pmatrix} 1 & 2-i & -3 \\ i & 0 & 1+i \end{pmatrix}.$$

Transposing does not conjugate complex entries: the entry i remains i.

Theorem 5.4. Every matrix satisfies $(A^{\mathsf{T}})^{\mathsf{T}} = A$.

Proof. The double transpose has the same size as A, and

$$\left((A^{\mathsf{T}})^{\mathsf{T}}\right)_{ij} = (A^{\mathsf{T}})_{ji} = a_{ij}$$

for every entry. Matrix equality follows. □

6 Special types of matrices

6.1 Square, zero, and identity matrices

Definition 6.1. A matrix is *square of order n* if it has size $n \times n$. Its *main diagonal* consists of $a_{11}, a_{22}, \ldots, a_{nn}$, in that order.

For instance, $\begin{pmatrix} 3 & -2 \\ -2 & 6 \end{pmatrix}$ is square of order two, with main diagonal $3, 6$. A 1×1 matrix is square of order one.

Definition 6.2 (Zero and identity matrices)**.** The *zero matrix* $0_{m\times n}$ has zero in every entry. The *identity matrix* of order n is

$$I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

Its entries are the *Kronecker delta*:

$$(I_n)_{ij} = \delta_{ij} = \begin{cases} 1 & i = j, \\ 0 & i \neq j. \end{cases}$$

Example 6.1.

$$0_{2\times 4} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad 0_{1\times 1} = (0).$$

A zero matrix need not be square. An identity matrix is square by definition. We sometimes write 0 or I when the dimensions are clear.

6.2 Diagonal, scalar, symmetric, and skew-symmetric matrices

Definition 6.3. Let A be square of order n.

1. A is *diagonal* if $a_{ij} = 0$ whenever $i \neq j$.
2. A is a *scalar matrix* if it is diagonal and every diagonal entry is the same scalar $\alpha \in \mathbb{F}$.
3. A is *symmetric* if $A^{\mathsf{T}} = A$, equivalently $a_{ij} = a_{ji}$ for all i, j.

4. A is *skew-symmetric* (also called antisymmetric) if $A^{\mathsf{T}} = -A$, equivalently $a_{ij} = -a_{ji}$ for all i, j.

Here $-A$ means the matrix whose entries are $-a_{ij}$.

Example 6.2 (Recognizing the definitions)**.** Over $\mathbb{R}$,

$$D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 4 \end{pmatrix}, \quad S = \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix}, \quad H = \begin{pmatrix} 3 & -2 \\ -2 & 6 \end{pmatrix}, \quad K = \begin{pmatrix} 0 & 4 \\ -4 & 0 \end{pmatrix}.$$

The matrix D is diagonal and symmetric, but not scalar. The matrix S is scalar, hence diagonal and symmetric. The matrix H is symmetric, but not diagonal. The matrix K is skew-symmetric.

Every identity matrix is scalar, diagonal, and symmetric. Every square zero matrix is scalar, diagonal, symmetric, and skew-symmetric. A nonsquare matrix is not in any of these square-matrix classes.

A scalar matrix has the form αI_n. Diagonal and symmetric matrices can have zero diagonal entries; the definitions do not require nonzero entries there.

Theorem 6.4 (Skew-symmetry when $1 + 1 \neq 0$)**.** Suppose the field $\mathbb{F}$ satisfies $1 + 1 \neq 0$.

1. Every diagonal entry of a skew-symmetric matrix is zero.
2. A matrix that is both symmetric and skew-symmetric is the zero matrix.

Proof. For a skew-symmetric matrix, $a_{ii} = -a_{ii}$, so $(1 + 1)a_{ii} = 0$. Since $1 + 1 \neq 0$ is invertible in the field, $a_{ii} = 0$. If A is both symmetric and skew-symmetric, then $a_{ij} = a_{ji} = -a_{ij}$ for all i, j. The same argument gives $a_{ij} = 0$ for every entry. □

Remark 6.5 (The field matters)**.** In $\mathbb{Z}_2$, $-1 = 1$, so $-A = A$ for every matrix. In particular,

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

is both symmetric and skew-symmetric over $\mathbb{Z}_2$, despite being nonzero and having nonzero diagonal entries. Thus the hypothesis $1 + 1 \neq 0$ cannot be dropped. Over a field with $1 + 1 = 0$, an identity matrix is skew-symmetric as well as symmetric.

6.3 Triangular matrices and row and column vectors

Definition 6.6. A square matrix A is *upper triangular* if $a_{ij} = 0$ whenever $i > j$, and *lower triangular* if $a_{ij} = 0$ whenever $i < j$.

Example 6.3.

$$U = \begin{pmatrix} 2 & -1 & 3 \\ 0 & 5 & 0 \\ 0 & 0 & -4 \end{pmatrix}, \qquad L = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 5 & 0 \\ -3 & 2 & -4 \end{pmatrix}$$

are upper triangular and lower triangular, respectively. Zeros are allowed on the main diagonal and on the permitted side of it. A matrix is both upper and lower triangular exactly when it is diagonal.

Theorem 6.7. A square matrix A is upper triangular if and only if A^{T} is lower triangular.

Proof. If $i < j$, then $(A^{\mathsf{T}})_{ij} = a_{ji}$ and $j > i$, so the relevant entry of A is below its diagonal. Thus upper triangularity of A gives lower triangularity of A^{T}. The converse follows by transposing again. □

Definition 6.8 (Row and column vectors)**.** A matrix of size $1 \times n$ is a *row vector*; a matrix of size $m \times 1$ is a *column vector*.

For example,

$$\begin{pmatrix} 2 & -1 & 0 & 5 \end{pmatrix} \quad \text{is a row vector, and} \quad \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} \quad \text{is a column vector.}$$

An $m \times n$ matrix has m rows, each a row vector of length n, and n columns, each a column vector of length m.

7 Scalar multiplication and matrix addition

7.1 Multiplying a matrix by a scalar

Definition 7.1. A *scalar* is an element of the field $\mathbb{F}$. If $\alpha \in \mathbb{F}$ and $A_{m\times n}$ is over $\mathbb{F}$, define αA entry by entry:

$$(\alpha A)_{ij} = \alpha a_{ij}.$$

The dimensions do not change.

Example 7.1 (The same array over two different fields)**.** Over $\mathbb{R}$,

$$4\begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix} = \begin{pmatrix} 4 & 8 \\ 12 & 20 \end{pmatrix}.$$

Over $\mathbb{Z}_7$, the corresponding calculation is

$$4\begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ 5 & 6 \end{pmatrix},$$

because the products are reduced modulo seven. The field is part of the calculation.

Example 7.2 (Complex scalars)**.**

$$(2-i)\begin{pmatrix} 1 & 0 & -1 \\ 2 & 3 & i \end{pmatrix} = \begin{pmatrix} 2-i & 0 & -2+i \\ 4-2i & 6-3i & 1+2i \end{pmatrix}.$$

For the final entry, $(2-i)i = 2i - i^2 = 1 + 2i$.

Theorem 7.2 (Rules for scalar multiplication)**.** For $\alpha, \beta \in \mathbb{F}$ and $A_{m\times n}$,

$$(\alpha\beta)A = \alpha(\beta A), \qquad (\alpha A)^{\mathsf{T}} = \alpha A^{\mathsf{T}},$$

$$1A = A, \qquad (-1)A = -A, \qquad 0A = 0_{m\times n}.$$

Proof. For the first identity, the (i,j) entries are $(\alpha\beta)a_{ij}$ and $\alpha(\beta a_{ij})$, equal by associativity in $\mathbb{F}$. For the transpose identity,

$$\big((\alpha A)^{\mathsf{T}}\big)_{ij} = (\alpha A)_{ji} = \alpha a_{ji} = (\alpha A^{\mathsf{T}})_{ij}.$$

The remaining statements follow from $1a = a$, $(-1)a = -a$, and $0a = 0$ in the field, applied to every entry. □

7.2 Addition and subtraction

Definition 7.3. If A and B are matrices of the same size over the same field, their sum and difference are defined by

$$(A+B)_{ij} = a_{ij} + b_{ij}, \qquad (A-B)_{ij} = a_{ij} - b_{ij}.$$

If the sizes do not agree, these operations are undefined.

Example 7.3. Over $\mathbb{C}$,

$$\begin{pmatrix} 2 & -1 \\ 0 & 3 \\ 4 & i \end{pmatrix} + \begin{pmatrix} -2 & 2 \\ i & 1 \\ 1 & -i \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ i & 4 \\ 5 & 0 \end{pmatrix}.$$

A 2×2 matrix cannot be added to a 2×3 matrix. Missing positions are not interpreted as zero: the sum is simply undefined.

Theorem 7.4 (Rules for matrix addition)**.** Let A, B, C have the same size $m \times n$ over $\mathbb{F}$, and let $\alpha, \beta \in \mathbb{F}$. Then

$$\begin{aligned} A + B &= B + A, & A + (B + C) &= (A + B) + C, \\ A + 0_{m\times n} &= A, & A + (-A) &= A - A = 0_{m\times n}, \\ \alpha(A + B) &= \alpha A + \alpha B, & (\alpha + \beta)A &= \alpha A + \beta A, \\ (A \pm B)^{\mathsf{T}} &= A^{\mathsf{T}} \pm B^{\mathsf{T}}. \end{aligned}$$

Proof. Each identity follows by checking the corresponding entries in $\mathbb{F}$. For example,

$$\big(\alpha(A+B)\big)_{ij} = \alpha(a_{ij} + b_{ij}) = \alpha a_{ij} + \alpha b_{ij} = (\alpha A + \alpha B)_{ij}.$$

Similarly,

$$\big((A+B)^{\mathsf{T}}\big)_{ij} = (A+B)_{ji} = a_{ji} + b_{ji} = (A^{\mathsf{T}} + B^{\mathsf{T}})_{ij}.$$

Use commutativity, associativity, additive identities and inverses, and distributivity in $\mathbb{F}$ for the other statements. □

Example 7.4 (Combining two measurement tables)**.** Suppose the rows represent two sensors and the columns represent three times. If

$$M = \begin{pmatrix} 10 & 12 & 11 \\ 7 & 8 & 9 \end{pmatrix}, \qquad E = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & -1 \end{pmatrix},$$

then adding the correction table gives

$$M + E = \begin{pmatrix} 11 & 11 & 11 \\ 7 & 10 & 8 \end{pmatrix}.$$

The positions must refer to the same sensors and times for this addition to represent the intended correction. This is a concrete reason to keep track of rows and columns.

8 Matrix multiplication and powers

8.1 A row times a column

Matrix multiplication is different from multiplying corresponding entries. We begin with a row vector followed by a column vector of the same length:

$$\begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix} \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix} = \sum_{p=1}^{n} v_p w_p.$$

Strictly, the result is a 1×1 matrix, which we usually identify with its scalar entry.

Example 8.1 (A weighted total)**.** If a purchase contains 2, 1, and 3 units of three items, priced at 4, 5, and 2 units of currency, respectively, its cost is

$$\begin{pmatrix}2 & 1 & 3\end{pmatrix}\begin{pmatrix}4\\5\\2\end{pmatrix} = 2\times 4+1\times 5+3\times 2 = 19.$$

The product of two 1×3 row vectors, in that order and shape, is undefined. Also, over $\mathbb{C}$ the multiplication defined here does not conjugate any entries; it is a matrix product, rather than the complex inner product.

8.2 The general product and its dimensions

Definition 8.1 (Matrix product)**.** Let $A_{m\times n}$ and $B_{n\times k}$ be matrices over the same field. The product AB has size $m\times k$, and

$$(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj} = \sum_{p=1}^{n} a_{ip}b_{pj}.$$

Thus entry (i,j) is row i of A multiplied by column j of B. The number of columns of A must equal the number of rows of B.

The dimension rule is

$$(m\times \underbrace{\quad n\quad}_{\text{must match}})(\underbrace{\quad n\quad}_{\text{must match}}\times k) \quad\longrightarrow\quad m\times k.$$

Example 8.2 (A rectangular product, entry by entry)**.** Let

$$A = \begin{pmatrix}2 & 1 & 0\\0 & 1 & 3\end{pmatrix}, \qquad B = \begin{pmatrix}4 & 1\\2 & 5\\1 & 2\end{pmatrix}.$$

The dimensions are $(2\times 3)(3\times 2)$, so AB is 2×2. Its entries are

$$\begin{aligned}(AB)_{11} &= 2\times 4+1\times 2+0\times 1 = 10,\\(AB)_{12} &= 2\times 1+1\times 5+0\times 2 = 7,\\(AB)_{21} &= 0\times 4+1\times 2+3\times 1 = 5,\\(AB)_{22} &= 0\times 1+1\times 5+3\times 2 = 11.\end{aligned}$$

Hence

$$AB = \begin{pmatrix}10 & 7\\5 & 11\end{pmatrix}.$$

Adding more rows to A or more columns to B preserves the matching inner dimension, although the output changes. Adding a column to A without adding a row to B breaks the dimension condition.

For general 2×2 matrices, the rule reads

$$\begin{pmatrix}a & b\\c & d\end{pmatrix}\begin{pmatrix}e & f\\g & h\end{pmatrix} = \begin{pmatrix}ae+bg & af+bh\\ce+dg & cf+dh\end{pmatrix}.$$

8.3 Order matters and zero products occur

Example 8.3 (Matrix multiplication is not commutative)**.** Let

$$P = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}, \qquad Q = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}.$$

Compute both products:

$$PQ = \begin{pmatrix} 2\times 1+(-1)\times 0 & 2\times 2+(-1)\times 1 \\ 1\times 1+3\times 0 & 1\times 2+3\times 1 \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 1 & 5 \end{pmatrix},$$
$$QP = \begin{pmatrix} 1\times 2+2\times 1 & 1\times(-1)+2\times 3 \\ 0\times 2+1\times 1 & 0\times(-1)+1\times 3 \end{pmatrix} = \begin{pmatrix} 4 & 5 \\ 1 & 3 \end{pmatrix}.$$

Therefore $PQ \neq QP$, even though both products are defined and have the same size. For rectangular matrices, both products may exist with different sizes, or only one product may be defined.

Example 8.4 (Why the order of two operations can matter)**.** Let

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad D = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}.$$

The matrix R rotates a real column vector by a quarter-turn, while D multiplies its first coordinate by 2 and its second by 3. Direct multiplication gives

$$RD = \begin{pmatrix} 0\times 2+(-1)\times 0 & 0\times 0+(-1)\times 3 \\ 1\times 2+0\times 0 & 1\times 0+0\times 3 \end{pmatrix} = \begin{pmatrix} 0 & -3 \\ 2 & 0 \end{pmatrix},$$

$$DR = \begin{pmatrix} 2\times 0+0\times 1 & 2\times(-1)+0\times 0 \\ 0\times 0+3\times 1 & 0\times(-1)+3\times 0 \end{pmatrix} = \begin{pmatrix} 0 & -2 \\ 3 & 0 \end{pmatrix}.$$

Applied to $(1,0)^{\mathsf{T}}$, these produce $(0,2)^{\mathsf{T}}$ and $(0,3)^{\mathsf{T}}$, respectively. In a product acting on a column vector, the matrix on the right acts first.

Example 8.5 (Nonzero matrices can have zero product)**.** Over $\mathbb{R}$, let $N = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}$. Then $N \neq 0$, but

$$N^2 = \begin{pmatrix} 0\times 0+2\times 0 & 0\times 2+2\times 0 \\ 0\times 0+0\times 0 & 0\times 2+0\times 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Thus nonzero square matrices can be zero divisors, and $AB = 0$ does not generally imply $A = 0$ or $B = 0$.

8.4 Powers of a square matrix

Definition 8.2. For a square matrix A of order n, define

$$A^0 = I_n, \qquad A^r = A^{r-1}A \quad (r \geq 1).$$

Thus $A^2 = AA$ and $A^3 = AAA$. Ordinary positive integer powers require a square matrix, because the repeated products must be defined.

Example 8.6. For $P = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}$,

$$P^2 = \begin{pmatrix} 2\times 2+(-1)\times 1 & 2\times(-1)+(-1)\times 3 \\ 1\times 2+3\times 1 & 1\times(-1)+3\times 3 \end{pmatrix} = \begin{pmatrix} 3 & -5 \\ 5 & 8 \end{pmatrix}.$$

Remark 8.3 (A familiar identity needs a condition)**.** Distributivity gives

$$(A+B)^2 = A^2 + AB + BA + B^2.$$

Replacing the middle two terms by $2AB$ is justified when $AB = BA$. In general, matrices do not commute, so the scalar identity cannot be used automatically. Likewise, the scalar binomial formula requires commutation when applied to matrices.

8.5 Rules for matrix multiplication

Theorem 8.4. All matrices below are over the same field, and all products and sums in each statement are assumed to have compatible dimensions.

1. Associativity: $(AB)C = A(BC)$.
2. Distributivity: $A(B+C) = AB + AC$ and $(A+B)C = AC + BC$.
3. For a scalar α, $\alpha(AB) = (\alpha A)B = A(\alpha B)$.
4. For $A_{m\times n}$, $I_m A = A I_n = A$.
5. $A_{m\times n} 0_{n\times k} = 0_{m\times k}$ and $0_{k\times m} A_{m\times n} = 0_{k\times n}$.
6. Transpose reverses the order: $(AB)^{\mathsf{T}} = B^{\mathsf{T}} A^{\mathsf{T}}$.

Before proving associativity, recall two rules for finite sums. A summation index is a dummy name:

$$\sum_{p=1}^{4} p^2 = \sum_{q=1}^{4} q^2 = 1 + 4 + 9 + 16 = 30.$$

Also, when the bounds are fixed and do not depend on the other index, finite sums may be interchanged:

$$\sum_{p=1}^{r} \sum_{q=1}^{s} c_{pq} = \sum_{q=1}^{s} \sum_{p=1}^{r} c_{pq}.$$

Both sides add exactly the same rs entries. For example,

$$\sum_{p=1}^{2} \sum_{q=1}^{3} pq = (1+2+3) + (2+4+6) = 18,$$

whereas the reversed order gives $(1+2) + (2+4) + (3+6) = 18$.

Proof. For associativity, let A be $m \times n$, B be $n \times r$, and C be $r \times k$. Both products have size $m \times k$. Their (i,j) entries satisfy

$$\begin{aligned}
\big((AB)C\big)_{ij} &= \sum_{q=1}^{r} (AB)_{iq} c_{qj} \\
&= \sum_{q=1}^{r} \left(\sum_{p=1}^{n} a_{ip} b_{pq} \right) c_{qj} \\
&= \sum_{q=1}^{r} \sum_{p=1}^{n} a_{ip} b_{pq} c_{qj} \\
&= \sum_{p=1}^{n} \sum_{q=1}^{r} a_{ip} b_{pq} c_{qj} \\
&= \sum_{p=1}^{n} a_{ip} \left(\sum_{q=1}^{r} b_{pq} c_{qj} \right) \\
&= \sum_{p=1}^{n} a_{ip} (BC)_{pj} = \big(A(BC)\big)_{ij}.
\end{aligned}$$

For distributivity, compute

$$\big(A(B+C)\big)_{ij} = \sum_p a_{ip}(b_{pj}+c_{pj}) = \sum_p a_{ip}b_{pj} + \sum_p a_{ip}c_{pj} = (AB+AC)_{ij}.$$

The proof for $(A+B)C$ is the same entry calculation on the other side. Scalar compatibility follows from

$$\alpha \sum_p a_{ip}b_{pj} = \sum_p (\alpha a_{ip})b_{pj} = \sum_p a_{ip}(\alpha b_{pj}).$$

For the identity matrix,

$$(I_m A)_{ij} = \sum_{p=1}^{m} \delta_{ip}a_{pj} = a_{ij}, \qquad (AI_n)_{ij} = \sum_{p=1}^{n} a_{ip}\delta_{pj} = a_{ij}.$$

For a product involving a zero matrix, every summand is zero, so every output entry is zero. Finally, if A is $m \times n$ and B is $n \times k$, both $(AB)^{\mathsf{T}}$ and $B^{\mathsf{T}}A^{\mathsf{T}}$ are $k \times m$, and

$$\begin{aligned}\big((AB)^{\mathsf{T}}\big)_{ij} &= (AB)_{ji} = \sum_{p=1}^{n} a_{jp}b_{pi} \\ &= \sum_{p=1}^{n} (B^{\mathsf{T}})_{ip}(A^{\mathsf{T}})_{pj} = (B^{\mathsf{T}}A^{\mathsf{T}})_{ij}.\end{aligned}$$

This proves the reverse-order rule. □